

Amy Wei

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Education

B.S.E. University of Michigan, Computer Science, Minor in Mathematics Ann Arbor, MI
Sept 2022 – present

- GPA: 3.9/4.0
- Relevant Courses: Software Engineering, Computer Security, Operating Systems, Machine Learning, Discrete Mathematics
- Selected Awards: Barry Goldwater Scholarship 2025

Publications

TestLoop: A Process Model Describing Human-in-the-Loop Software Test Suite Generation Aug 2025

Matthew C. Davis, Sangheon Choi, **Amy Wei**, Sam Estep, Brad A. Myers, Joshua Sunshine
<https://doi.org/10.1145/3765754> [🔗](#) (ACM Transactions on Software Engineering and Methodology)

TerzoN: Human-in-the-Loop Software Testing with a Composite Oracle Apr 2025

Matthew C. Davis, **Amy Wei**, Brad A. Myers, Joshua Sunshine
<https://doi.org/10.1145/3729359> [🔗](#) (ACM Foundations of Software Engineering)

Manuscripts Under Submission

Sweat the Details: Exercise Breaks and AI Assistance in Code Summarization

Amy Wei, Priscila Santiesteban, Westley Weimer
 IEEE Transactions on Software Engineering (*under submission*)

Research Experience

Software and Cognition Lab, University of Michigan, Undergraduate Researcher Ann Arbor, MI
Nov 2024 – present

- Advised by Dr. Westley Weimer and PhD student Priscila Santiesteban
- Developed paper ideas by performing literature review of 18 papers
- Designed source code stimuli and programming tasks for human study
- Conducted study of 47 participants investigating effect of exercise breaks and AI assistance on code summarization performance
- Used qualitative analysis and mixed effects linear regression to analyze results
- Wrote and submitted manuscript to IEEE Transactions on Software Engineering

REU in Software Engineering '24, Carnegie Mellon University, Research Assistant Pittsburgh, PA
May 2024 – Aug 2024

- Advised by Dr. Joshua Sunshine, Dr. Brad Myers, and PhD student Matthew Davis
- Built TerzoN, a usable testing tool for TypeScript, during 10-week program sponsored by NSF, Amazon, and Meta
- Implemented support for 3 types of testing (fuzzing, example-based testing, property-based testing)
- Designed user study with 14 participants comparing the tool NaNoFuzz with the traditional property-based testing tool fast-check to investigate the effect of modality on bug-finding performance and usability

REU in Software Engineering '23, Carnegie Mellon University, Research Assistant Pittsburgh, PA
May 2023 – Aug 2023

- Advised by Dr. Joshua Sunshine, Dr. Brad Myers, and PhD student Matthew Davis
- Performed heuristic evaluations of 5 testing tools as benchmark (Hypothesis, Jest, Randoop, JQF, Evosuite)

- Implemented sorting, human annotation, and validator function features to NaNoFuzz to support multiple modes of testing
- Presented at weekly meetings and shared findings in department-wide poster session with 100+ attendees

Teaching

Discrete Mathematics (EECS 203) , Instructional Aide & Homework Czar	Ann Arbor, MI Jan 2024 – present
• Served as a TA for 4 semesters and led homework management team as czar • Taught discussion section of 40+ students and held weekly office hours to support student learning • Hired and managed team of 16 graders to ensure that students received accurate and timely feedback • Published homework grades for 600+ students and led end-to-end homework duties • Developed supplemental module of 33 problems to help students build mathematical skills for writing proofs	
Elementary Programming Concepts (EECS 183) & Linear Algebra (MATH 214) , Tutor	Ann Arbor, MI Sept 2023 – Jan 2024
• Met weekly with students through U-M CSE Tutoring to review course concepts and reinforce learning	

Honors & Awards

Barry Goldwater Scholarship, \$7,500 , Barry Goldwater Foundation	2025
The preeminent undergraduate award in STEM. Awarded for excellence in research, academics, and contributions to the community. 50 scholars in mathematics and computer science were selected nationally in 2025.	
College of Engineering Scholarship Award, \$3,000 , U-M College of Engineering	2025
Arlen R. Hellwarth Award, \$500 , U-M College of Engineering	2025
College of Engineering Scholarship Award, \$2,000 , U-M College of Engineering	2024
Distinguished Leadership Award , U-M College of Engineering	2024
James B. Angell Scholar , University of Michigan	2024
George Landes Prize, Technical Communication First Year, \$150 , University of Michigan	2024
Engineering Honors Program , University of Michigan	2023-2026
Dean's Honor List and University Honors , University of Michigan, awarded all semesters	2023-2025
William J. Branstrom Freshman Prize , University of Michigan	2023

Activities

IEEE-Eta Kappa Nu (EECS Honor Society)	Sept 2023 – present
• Recognized for earning the highest number of service hours in University of Michigan chapter	
Applied Linear Algebra (MATH 214), Grader	Sept 2023 – Jan 2024
• Graded weekly homework for 300+ students to provide timely feedback • Drafted comprehensive grading rubrics and wrote individualized comments to support student learning	
Campus Symphony Orchestra	Sept 2022 – Jan 2023
• Selected to rehearse weekly and perform in university-wide concerts each semester as Violin I and Principal Violin II	
Society of Women Engineers	Sept 2022 – present
• Volunteered at Elementary Engineering Exploration Day to introduce young students to STEM • Held a charity Fun Run in support of University of Michigan Health C.S. Mott Children's Hospital	
Women's Glee Club Choir	Sept 2022 – Apr 2023

- Selected to rehearse weekly and perform in university-wide concerts each semester as Alto I

Skills

Technical: C/C++, Python, JavaScript/TypeScript, Java, R, HTML/CSS, Git, LaTeX